Charlene Teo
Mobile: +65 90976711 | Email: charleneteoen10@gmail.com | Linkedin: Charlene Teo | Portfolio: Charlene's Portfolio

***ACADEMIC QUALIFICATIONS*:**

Singapore University Of Technology and Design — Sep 2021 - May 2025
Bachelors of Science, Design and Artificial Intelligence
Computer Science and Design minor

Chalmers University of Technology — Jan 2024 - Jun 2024
Global Spring Exchange Programme

***WORK EXPERIENCES*:**

United Overseas Bank Analyst — Aug 2025 - Present
-Built an end-to-end document processing pipeline (classification → OCR/LLM extraction → rule-based validation), reducing manual processing workflows by 80%
-Implemented LLM-based document type identification with prompt engineering to classify uploaded documents and route them to the right processing workflow
-Benchmarked 4 on-premise vision and language models (Qwen3.5, PaddleOCR-VL, GLM, DeepSeek-V4) on 210 financial documents for field-level accuracy and processing speed, selecting the setup that reached 90% extraction accuracy with few-shot prompting and JSON schema validation

***INTERNSHIP*:**

Deutsche Bank Fall Analyst — Aug 2023 - Dec 2023
-Collaborated with regional partners to gather specific requirements for FX trade bookings across APAC and developed automation of trade bookings which included parsing of trade data, validating it for accuracy and generating booking entries with an FTE saving of 0.313
-Automated the FX email confirmation system using Python to send remittance details to 300+ clients monthly, reducing manual processing effort by 15%

***PROJECTS*:**

AML Transaction Monitoring Platform
-Built an Airflow pipeline ingesting 20,000 synthetic bank transactions per batch with automated validation, idempotent loads and quarantine of invalid batches, supported by over 60 automated tests and GitHub Actions CI
-Trained scikit-learn laundering-detection models tracked in MLflow, promoting a new model only when it outperforms the current one, and served them through a FastAPI scoring API that shares feature code with training
-Modelled transactions as a Neo4j graph, using Cypher queries to detect 127 money-laundering cycles at 100% precision and adding graph features that raised the model's average precision from 0.18 to 0.42
-Developed a React and TypeScript analyst dashboard whose alert queue achieves a 43% laundering hit rate against a 1% base rate, deployable with Docker and Kubernetes.

Predictive Maintenance of Military Vehicles
-Collaborated with ST Engineering to develop an AI-powered predictive maintenance system for military vehicles, improving combat readiness through early fault detection and proactive maintenance scheduling
-Built an ensemble unsupervised learning model achieving 89% accuracy on unlabeled time series sensor data, combining statistical methods, clustering algorithms(K-means, DBSSCAN), and deep learning techniques to identify anomalies
-Developed interactive dashboards for technicians, drivers, and administrators to monitor fleet health, report issues, and detect potential failures before they occur, optimizing maintenance workflows and reducing unplanned downtime

Seamless Banking Web Application for Foreigners
-Collaborated with DBS to develop a full-stack web banking application that streamlined online account opening for foreigners without Singpass access, reducing account opening time from 2 weeks to 3 days
-Integrated OCR-based document scanning to extract key information from passports, work passes, and proof-of-address documents, reducing manual input errors and verification time
-Implemented facial recognition and NFC-based passport authentication to verify user identity and document authenticity

***HACKATHON*:**

JP Morgan Code For Good
-Developed a web application connecting urban gardeners and promoting food security through community engagement
-Implemented a plant recommendation survey, condition-based plant search, and peer-to-peer produce barter system
-Integrated social features including community forum links, interactive mapping of planting areas, and environmental condition filters to support beginner gardeners

SKILLS AND INTEREST:

Languages: Python, JavaScript, Ruby, SQL
Cloud & DevOps: Git, Docker, AWS, CI/CD pipelines, Google Cloud Platform
Certifications: Claude with Amazon Bedrock, Introduction to Model Context Protocol, Model Context Protocol: Advanced Topics, AI Fluency: Framework and Foundations, AI Capabilities and Limitations